

Stealing Professor Miller's Vegan Sludge

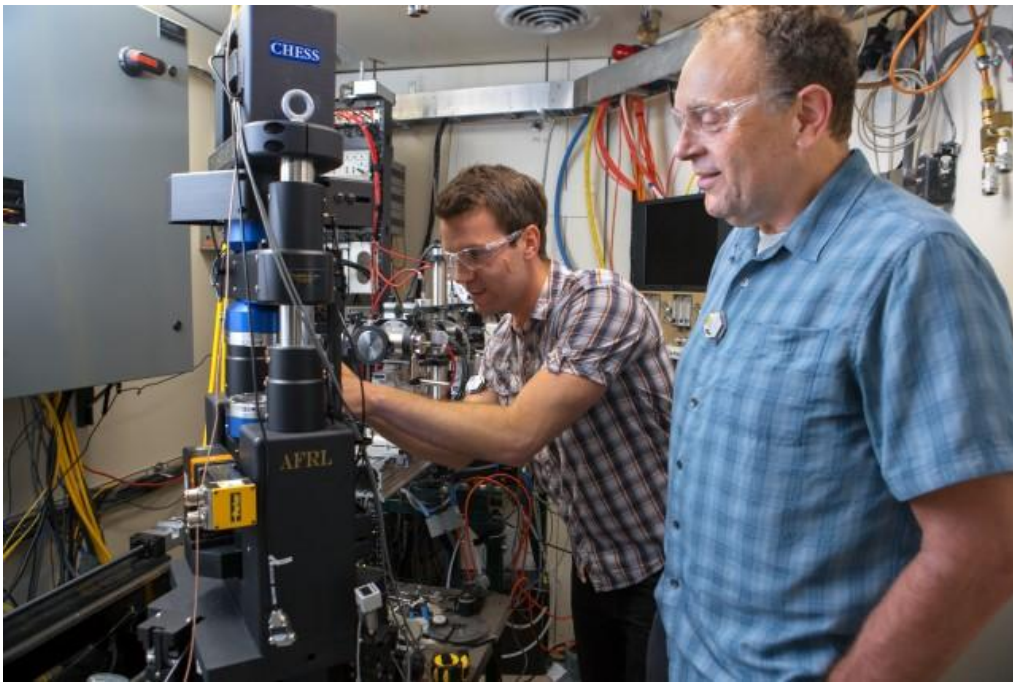
Academia is filled with life-long nerds like me; I started writing my first intro fluid mechanics textbook at the age of 17 during a period of friendlessness and social marginalization. I was such a prototypical nerd I was even bullied by my high school math teachers (true story). I remember the bullies when I was a kid, they looked kind of like #71 on the right.



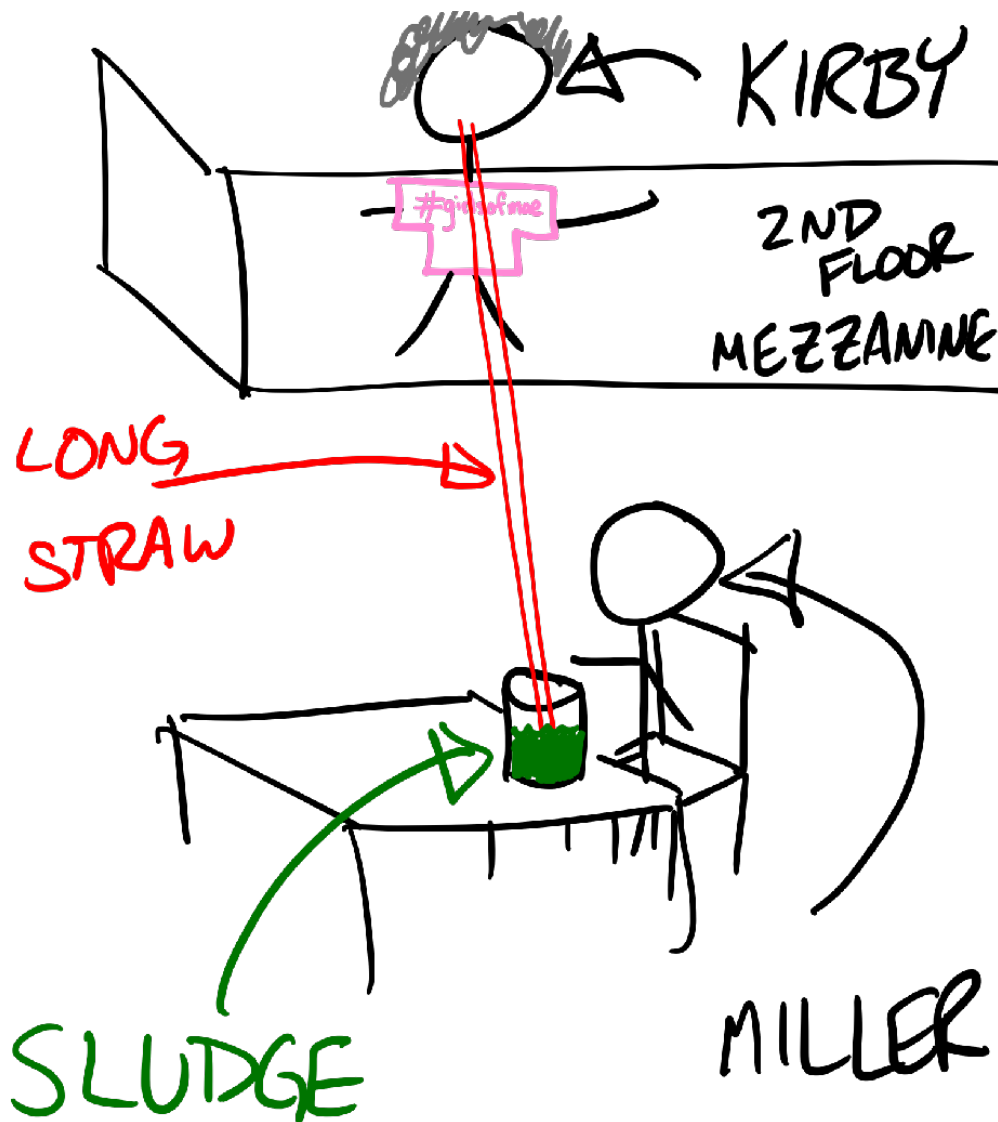
Academia does not have many former NFL offensive linemen. How many people have the skills to be an NFL player and the skills to be an Ivy League professor? Not sure, but Professor Miller does. That picture above is when he was an all-American in college; below is a picture of him in the NFL (#71 in the background):



And here he is at the Cornell High Energy Synchrotron Source:



Anyway, he used to be this big gruff football player but now Miller wears sweater vests, is vegan, and drinks herbal tea and kale smoothies. Which brings us to the fluid mechanics. Miller hangs out in Duffield sometimes drinking his kale smoothie thing in a big YETI travel mug. Let's suppose Professor Kirby decides to steal the kale sludge from Professor Miller while Professor Miller is eating lunch in Duffield Hall and isn't looking. Professor Kirby proposes to do this by using a long rigid metal straw extending from the balcony on the 2nd floor mezzanine (4 meters up). Here is a diagram of what Professor Kirby has in mind:



Does this work according to the hydrostatic equation? Answer by calculating what pressure Prof. K's mouth/lungs would have to make if the sludge filled the straw to a height equal to the 2nd floor mezzanine. Here we postulate that sludge fills the straw, and see if the resulting pressure values are plausible. If implausible, we need to explain what happens by showing what is wrong with a hydrostatic equation analysis. Also, if it works on the 2nd floor mezzanine, Prof Kirby is thinking about trying on the 4th floor mezzanine.

Relevant Equations

The hydrostatic equation is

$$\frac{\partial p}{\partial z} = -\rho g.$$

We need to know the density of vegan sludge. Water's density is 1000 kg/m³; I estimate vegan sludge to be about 1200 kg/m³.

Boundary Conditions

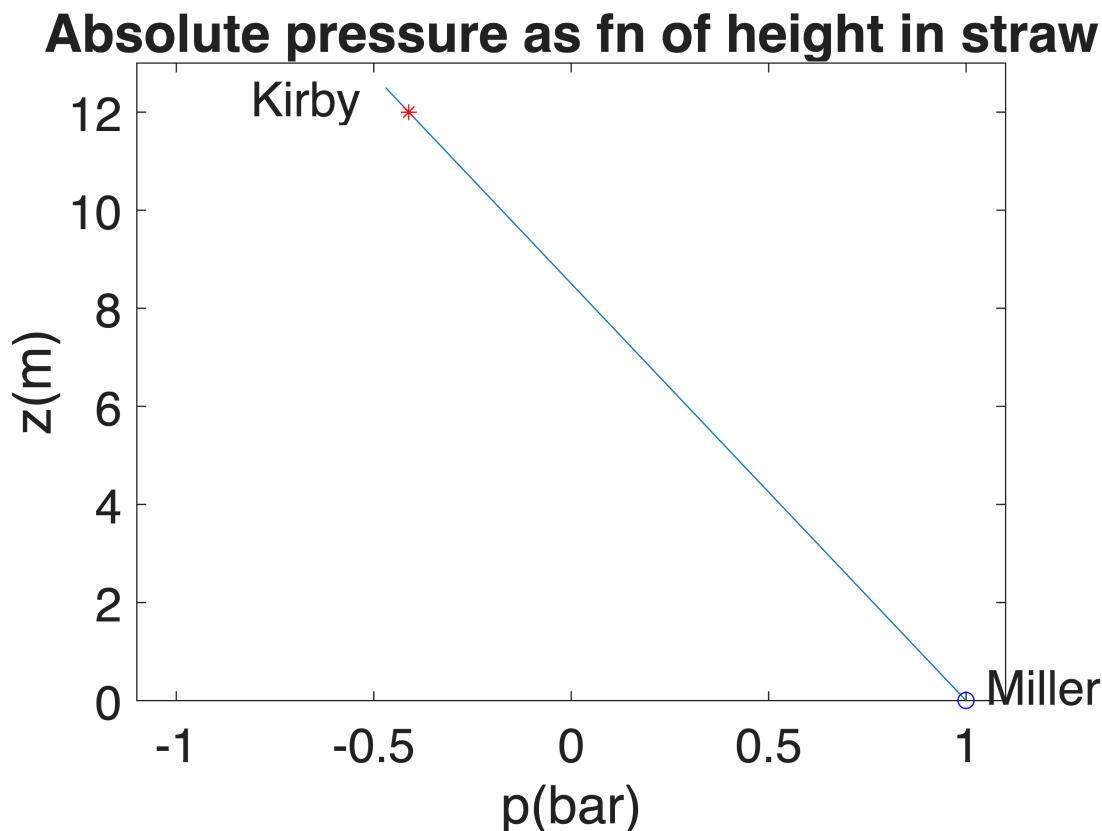
We know that the pressure at the surface of Miller's cup of sludge is the atmospheric pressure, approx $p = 0.1$ MPa.

Solution

We can integrate the differential equation and get

$$p(z) = p(0) - \rho g z = (1e5) - (1200)(9.8)z,$$

which is used in the MATLAB code below. The drop-down menu can be used to place Prof. Kirby (a) on the atrium floor (b) on the 2nd-floor mezzanine, and (c) on the 4th-floor mezzanine.



Kirby's elevation in meters is: 12

The absolute pressure (in bar) Kirby needs to generate
to pull sludge to this height is: -0.4112

The blue circle labels the pressure at Miller's cup of sludge. The red star labels the pressure at Professor Kirby, where he is trying to generate a low pressure to suck up the sludge. The pressure required to pull the sludge up to the 4th-floor mezzanine is a NEGATIVE absolute pressure whereas the pressure required to pull the sludge up to the 2nd-floor mezzanine is about half of an atmosphere.

Interpretation

Under normal, macroscopic conditions, fluids exist only at positive pressures. Gases have very little intermolecular forces; there is no way for them to support a tension i.e., a negative absolute pressure. They can support compression by colliding with each other but that's it. Liquids have intermolecular forces, but if they are at equilibrium with vapor, as soon as the absolute pressure gets low enough, they will cavitate or boil. So liquids can have negative pressure, but only if they are kept out of equilibrium.

It is perfectly realistic for Professor Kirby to generate an absolute pressure of about 1/2 bar with his mouth, so this should work on the 2nd floor mezzanine (you can create 1/10 bar of vacuum with your lungs, and about 2/3 bar of vacuum with your mouth).

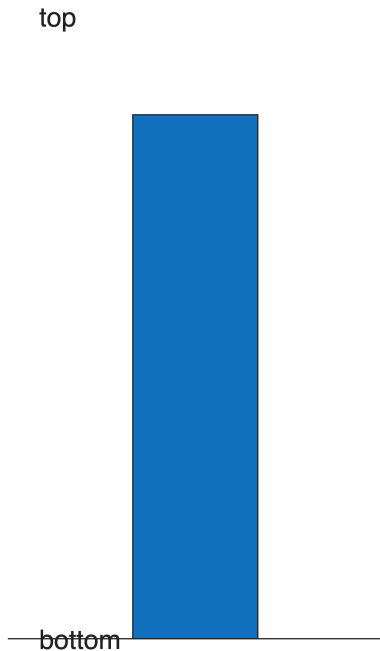
However, there is no way for water or vegan sludge to exist at -0.4 bar absolute; this would imply that the sludge is in tension. This is not possible to achieve, because the sludge would boil. The hydrostatic equation predicts what would happen if the straw were filled with fluid, but that situation is not possible to achieve if Professor Kirby sucks from the 4th floor mezzanine. He can't pull the sludge up the straw.

From the 4th floor mezzanine, if a vacuum is strong enough, it always generates boiling. If relatively weak, it will raise the water only a small amount and the water will be at equilibrium with its vapor. How high the water can be raised is sensitive to temperature because temperature dictates the vapor pressure.

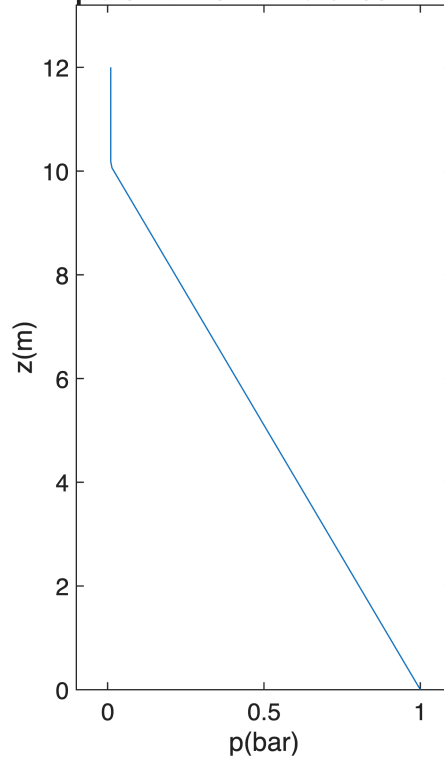
What happens when the hydrostatic equation doesn't describe the whole problem

The hydrostatic equation describes motionless, single-phase fluid in a gravitational field. When the solution that the hydrostatic equation provides contradicts some other aspect of the problem, then the fluid must not be motionless or must not be single-phase... or something. Below we can change the temperature (which changes the saturation pressure of water) and the pressure applied to the top of the straw when we try to suck the sludge up to the 4th floor mezzanine. The figure below shows at left a visualization of the column of water which rises and lowers in response to inputs, and at right the pressure distribution we observe. When the pump pressure tries to pull the water below its saturation value, we see a pressure discontinuity associated with nonequilibrium boiling at the surface. That nonequilibrium is not described by our fluid mechanics equations; it is described by water's equation of state (or, I guess, sludge's equation of state). This means that how high we can suck up water (or sludge) is a function of its temperature.

visualization of water column



p distribution in water column



The saturation pressure of water in bar is: 0.009

The pressure the pump pulls in bar is: 0.01

The pressure at the top in bar is: 0.01

The height of the water column in m is: 10.1024

Followup Problems

Answering these questions will require knowledge of the equilibrium properties of water. Consider water instead of sludge so that you can look up these equilibrium properties.

1. Consider water at room temperature (25C). Assume Professor Kirby has a very strong vacuum pump. How high can Professor Kirby suck up the water from Professor Miller's cup? Graph the pressure distribution in the water as a function of z .
2. Does the height that Professor Kirby can pull up the water change if the temperature of the cup, water, and straw are changed to 70C? How about 5C? Graph the pressure distribution in the water as a function of z in both cases.
3. Now consider a different configuration. Suppose the straw extends to the 4th floor mezzanine. The straw is **sealed** at the bottom and open at the top, and no vacuum is present. Can the straw be filled by pouring water into it from the top? If so, what is the pressure distribution in the straw?
4. Assume a filled straw extending to the 4th floor mezzanine is sealed at both ends. What can be specified about the pressure distribution inside the straw?
5. Assume the filled straw extending to the 4th floor mezzanine starts out full and sealed at both ends. Then it develops a leak at the bottom and water leaks out until the system comes to equilibrium. If the fluid filling the

straw is uniform at each height z , where in the straw will we find water when equilibrium is reached? What will the pressure distribution be? How much water will have leaked out?

6. Can a positive-pressure pump at the base of the straw push water higher up than a vacuum pump at the top? Or not as high? Or the same height? Why?